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FUEL TREATMENT SYSTEM FOR HYBRID GAS-ELECTRIC PROPULSION USING HYDROCARBON FUEL

Abstract

A fuel treatment system for a hybrid gas-electric propulsion system using a hydrocarbon fuel includes a fuel pre-treatment unit, a recuperator including a first fuel passage and a second fuel passage where the first fuel passage is in fluid communication with a fuel outlet of the fuel pre-treatment unit. A partial oxidation reformer includes a heated fuel inlet and a reformed fuel outlet. The heated fuel inlet is in fluid communication with the fuel pre-treatment unit via the first fuel passage. A solid oxide fuel cell includes an anode inlet and an anode outlet. The anode inlet is in fluid communication with the reformed fuel outlet of the partial oxidation reformer, and the anode outlet is in fluid communication with the second fuel passage of the recuperator. A combustor is in fluid communication with the anode outlet via the second fuel passage.

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Background/Summary

FIELD

[0001] The present disclosure relates to a hybrid gas-electric propulsion system, and more particularly to a fuel treatment system for a hybrid gas-electric propulsion system using a hydrocarbon fuel.

BACKGROUND

[0002] A hybrid gas-electric propulsion system depends on electrical power that may be generated by processing hydrogen fuel during operation to power a fan section or other components of the propulsion system. Carrying hydrogen fuel onboard an aircraft requires bulky fuel tanks and cooling systems, thus adding weight to the hybrid gas-electric propulsion system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

[0004] FIG. **1** is a perspective view of an exemplary aircraft that may incorporate at least one exemplary embodiment of the present disclosure.

[0005] FIG. **2** is a schematic cross-sectional view of an exemplary turbofan engine as may be incorporated into a propulsion system as shown in FIG. **1**, in accordance with an exemplary embodiment of the present disclosure.

[0006] FIG. **3** is a schematic diagram of a hybrid gas-electric propulsion system for an aircraft as shown in FIG. **1**, or other vehicle, in accordance with an exemplary embodiment of the present disclosure.

[0007] FIG. **4** is a schematic diagram of a fuel treatment system for the hybrid gas-electric propulsion system as shown in FIG. **3**, according to exemplary embodiments of the present disclosure.

[0008] FIG. **5** is a schematic diagram of a portion of the fuel treatment system for the hybrid gas-electric propulsion system as shown in FIG. **4**, with the addition of a post-heat heat exchanger, according to an exemplary embodiment of the present disclosure.

DETAILED DESCRIPTION

[0009] Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.

[0010] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

[0011] The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise. The term “at least one of” in the context of, e.g., “at least one of A, B, and C” refers to only A, only B, only C, or any combination of A, B, and C.

[0012] As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. Furthermore, the terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

[0013] The term “turbomachine” refers to a machine including one or more compressors, a heat generating section (e.g., a combustion section), and one or more turbines that together generate a torque output. The term “gas turbine engine” refers to an engine having a turbomachine as all or a portion of its power source. Example gas turbine engines include turbofan engines, turboprop engines, turbojet engines, turboshaft engines, etc., as well as hybrid-electric versions of one or more of these engines.

[0014] The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.

[0015] The phrases “from X to Y” and “between X and Y” each refers to a range of values inclusive of the endpoints (i.e., refers to a range of values that includes both X and Y).

[0016] The terms “fluidly connected” and “fluidly coupled” refer to a fluid connection made between two or more components or systems via pipes, fluid couplings, conduits, valves, and the like, such that a fluid (liquid or gas) may flow therebetween. The term “fluid communication” implies that a fluid (liquid or gas) flows between two or more components or systems. The term “thermal communication” implies that two components, systems, or features are formed, positioned, or configured to transfer thermal energy or heat therebetween.

[0017] The present disclosure is generally related to a hybrid-electric propulsion system which uses hydrocarbon fuel. The system disclosed herein provides a near-term solution towards an aviation goal of achieving zero or near-zero emissions via electrification using less bulky fuel tanks compared to Liquid Hydrogen (LH2) fuel-based systems.

[0018] Generation of substantial electric power using hydrocarbon fuel is a challenge in hybrid-electric engines due to coke formation in the fuel at the high operating temperature needed by the efficient electric power generation systems. This disclosure introduces a fuel treatment system for processing hydrocarbon fuel in a hybrid-electric aircraft by incorporating solid-oxide fuel cell with a partial oxidation reformer. Solid oxide fuel cells (SOFC) can generate large amounts of power (˜100 MW) due to their high operating temperature (˜800 C) and high efficiency, which makes the suitable for hybrid-electric aircraft engines. Also, SOFC is not vulnerable to carbon-monoxide (CO) poisoning and does not require expensive catalysts due to its high temperature—making it low cost. A partial oxidation reformer is used to convert the hydrocarbon fuel to a hydrogen fuel H2, which is needed by the SOFC. Since the reaction in both units is exothermic, the incorporated system is self-sustaining after cold start-up. The high fuel inlet temperature needed by the SOFC (˜800 C) is supplied by the partial oxidation reformer while the SOFC also acts as a water-gas shift reactor by converting the CO produced by the reformer into CO2 using water produced in the SOFC, which subsequently process more hydrogen fuel to increase electric power generated.

[0019] This disclosure introduces a fuel treatment system for a hybrid-electric propulsion using hydrocarbon fuel. The system includes a fuel pre-treatment unit, a recuperation heat exchanger, a partial oxidation reformer, a solid-oxide fuel cell, and a combustion chamber that enable combustion of both hydrocarbon fuel and hydrogen. In exemplary configurations the fuel pre-treatment unit deoxygenates and preheats the hydrocarbon fuel. The fuel pre-treatment unit may include a fuel-air heat exchanger, a fuel deoxygenator or “deoxygenation unit”, and may include, a fuel-oil heat exchanger.

[0020] Referring now to the drawings, FIG. **1** is a perspective view of an exemplary aircraft **10** that may incorporate at least one exemplary embodiment of the present disclosure. As shown in FIG. **1**, aircraft **10** has a fuselage **12**, wing(s) **14** attached to the fuselage **12**, and an empennage **16**. The aircraft **10** further includes a propulsion system **18** that produces a propulsive thrust to propel the aircraft **10** in flight, during taxiing operations, etc. Although the propulsion system **18** is shown attached to the wing(s) **14**, in other embodiments it may additionally or alternatively include one or more aspects coupled to other parts of the aircraft **10**, such as, for example, the empennage **16**, the fuselage **12**, or both. The propulsion system **18** includes at least one gas turbine engine **20** (two gas turbine engines shown). Each gas turbine engine **20** is mounted to aircraft **10** in an under-wing configuration. Each gas turbine engine **20** is capable of selectively generating propulsive thrust for the aircraft **10**. Each gas turbine engine **20** may be configured to burn various forms of fuel including, but not limited to unless otherwise provided, jet fuel/aviation turbine fuel, and hydrogen fuel. In addition, or in the alternative, each gas turbine engine **20** may be at least partially operated using onboard generated electric power.

[0021] As further show in hidden lines in FIG. **1**, the aircraft **10** may include a cabin **22** for carrying passengers or cargo, an environmental control system **24** for supplying air and for controlling temperatures onboard the aircraft **10** and may include an oxygen storage tank **26** for use with the environmental control system or for other uses onboard the aircraft **10**. In particular embodiments, the aircraft **10** may include a vent **28** for venting air or exhaust to the atmosphere.

[0022] FIG. **2** is a schematic cross-sectional view of a turbofan engine **120** as may be incorporated into the propulsion system **18** shown in FIG. **1**, in accordance with an exemplary embodiment of the present disclosure. More particularly. for the embodiment of FIG. **2**, the turbofan engine **120** is a high-bypass gas-electric (hybrid) turbofan jet engine. As shown in FIG. **2**, the turbofan engine **120** defines an axial direction A (extending parallel to a longitudinal centerline **122** provided for reference) and a radial direction R. In general, the turbofan engine **120** includes a fan section **124** and a core turbine engine **126** disposed downstream from the fan section **124**.

[0023] The core turbine engine **126** depicted generally includes an outer casing **128** that is substantially tubular and that defines an annular inlet **130**. The outer casing **128** encases, in serial flow relationship, a compressor section including a booster or low pressure (LP) compressor or “LP compressor **132**” and a high pressure (HP) compressor or “HP compressor **134**”, a combustion section **136**, a turbine section including a high pressure (HP) turbine or “HP turbine **138**” and a low pressure (LP) turbine or “LP turbine **140**”, and a jet exhaust nozzle section **142**. A high pressure (HP) shaft or spool or “HP shaft or spool **144**” drivingly connects the HP turbine **138** to the HP compressor **134**. A low pressure (LP) shaft or spool or “LP shaft or spool **146**” drivingly connects the IP turbine **140** to the LP compressor **132**. The LP compressor **132**, HP compressor **134**, combustion section **136**. HP turbine **138**, LP turbine **140**, and the jet exhaust nozzle section **142** together define, in serial flow order, a core air flow path through the turbofan engine **120**.

[0024] For the embodiment depicted, the fan section **124** includes a fan **148** having a plurality of fan blades **150** coupled to a disk **152** in a circumferentially spaced apart manner. As depicted, the fan blades **150** extend outwardly from disk **152** generally along the radial direction R. Each fan blade **150** is rotatable relative to the disk **152** about a pitch axis P by virtue of the fan blades **150** being operatively coupled to a pitch change mechanism **154** configured to collectively vary the pitch of the fan blades **150** in unison. The fan blades **150**, disk **152**, and pitch change mechanism **154** are together rotatable about the longitudinal centerline **122** by LP shaft or spool **146** across a power gearbox **156**. Power gearbox **156** includes a plurality of gears for adjusting the rotational speed of the fan **148** relative to the LP shaft or spool **146** to a more efficient rotational fan speed.

[0025] Referring still to the exemplary embodiment of FIG. **2**, disk **152** is covered by a front hub **158** that is rotatable and aerodynamically contoured to promote an airflow across and through the plurality of fan blades **150**. Additionally, the fan section **124** includes an annular fan casing or nacelle **160** that circumferentially surrounds the fan **148** and/or at least a portion of the core turbine engine **126**. The nacelle **160** is supported relative to the core turbine engine **126** by a plurality of circumferentially spaced outlet guide vanes **162**. Moreover, a downstream section **164** of the nacelle **160** extends over an outer portion of the core turbine engine **126** to define a bypass airflow passage **166** therebetween.

[0026] During operation of the turbofan engine **120**, air **168** enters the turbofan engine **120** through

an associated inlet **170** of the nacelle **160** and/or fan section **124**. As the air **168** passes across the fan blades **150**, a first portion **172** of air **168** is directed or routed into the bypass airflow passage **166** and a second portion **174** of air **168** is directed or routed into a core air flowpath **147**, or more specifically into the LP compressor **132**. The ratio between the first portion **172** of air **168** and the second portion **174** of air **168** is commonly known as a bypass ratio. For the exemplary embodiment depicted, the bypass ratio may be at least about 8:1. Accordingly, the turbofan engine **120** may be referred to as an ultra-high bypass turbofan engine. The pressure of the second portion **174** of air **168** is then increased as it is routed through the HP compressor **134** and into the combustion section **136**, where it is mixed with fuel and burned to provide combustion gases **176**.
[0027] The combustion gases **176** are routed through the HP turbine **138** where a portion of thermal and/or kinetic energy from the combustion gases **176** is extracted via sequential stages of HP turbine stator vanes **178** that are coupled to the outer casing **128** and HP turbine rotor blades **180** that are coupled to the HP shaft or spool **144**, thus causing the HP shaft or spool **144** to rotate, which supports operation of the HP compressor **134**. The combustion gases **176** are then routed through the LP turbine **140** where a second portion of thermal and kinetic energy is extracted from the combustion gases **176** via sequential stages of LP turbine stator vanes **182** that are coupled to the outer casing **128** and LP turbine rotor blades **184** that are coupled to the LP shaft or spool **146**, which causes the LP shaft or spool **146** to rotate, which supports operation of the LP compressor **132** and/or rotation of the fan **148**.
[0028] The combustion gases **176** are subsequently routed through the jet exhaust nozzle section **142** of the core turbine engine **126** to provide propulsive thrust. Simultaneously, the pressure of the first portion **172** of air **168** is substantially increased as the first portion **172** of air **168** is routed through the bypass airflow passage **166** before it is exhausted from a fan nozzle exhaust section **186** of the turbofan engine **120**, also providing propulsive thrust. The HP turbine **138**, the LP turbine **140**, and the jet exhaust nozzle section **142** at least partially define a hot gas path **188** for routing the combustion gases **176** through the core turbine engine **126**.
[0029] Turbofan engine **120** depicted in FIG. **2** is configured as an aeronautical gas turbine engine. Aeronautical gas turbine engines, as compared to land-based gas turbine engines, are designed to maximize power output and efficiency while minimizing an overall weight of the gas turbine engine itself, as well as any required accessory systems. It should be appreciated, however, that the turbofan engine **120** depicted in FIG. **2** is provided by way of example only, and that in other exemplary embodiments, the turbofan engine **120** may have any other suitable configuration. It should also be appreciated that in still other exemplary embodiments, aspects of the present disclosure may be incorporated into any other suitable gas turbine engine. For example, in other exemplary embodiments, aspects of the present disclosure may be incorporated into, e.g., a turboprop engine, a turboshaft engine, or a turbojet engine.
[0030] Referring now to FIG. **3**, is a schematic diagram of a hybrid gas-electric propulsion system **200** for aircraft **10** shown in FIG. **1**, or other vehicle in accordance with an exemplary embodiment of the present disclosure. For the embodiment depicted, the hybrid gas-electric propulsion system **200** includes a fuel tank **202** fluidly coupled to a fuel treatment system **300** for treating a hydrocarbon fuel (**500**) in the hybrid gas-electric propulsion system **200**. Fuel tank **202** is configured to contain and provide hydrocarbon fuel **500** to the fuel treatment system **300**. In particular embodiments, a fuel pump **204** provides a motive force for moving the hydrocarbon fuel **500** from the fuel tank **202** to and at least partially through the fuel treatment system **300**.
[0031] In exemplary embodiments, the fuel treatment system **300** is fluidly coupled to the combustion section **136** of the turbofan engine **120** shown schematically in FIG. **2**. As shown in FIG. **3**, the fuel treatment system **300** may be electrically connected to an electric machine **206** such as but not limited to an electric motor. In addition, or in the alternative, the fuel treatment system **300** may be electrically connected to an auxiliary electrical system **208** of the aircraft **10** (FIG. **1**) such as but not limited to the environmental control system **24** (FIG. **1**) or a flight control

system (not shown).

[0032] In exemplary embodiments, as shown in FIG. **3**, electric machine **206** is mechanically coupled via a shaft **210** to a fan gearbox **212**. The fan gearbox **212** may be coupled to the fan section **124** of the turbofan engine **120** via a fan shaft **214**. In particular configurations, the fan gearbox **212** may be mechanically coupled to the power gearbox **156**. In operation, the electric machine **206** receives electrical power **216** from the fuel treatment system **300**. converts the electrical power **216** into rotational or mechanical power to turn shaft **210** and to drive the fan gearbox **212**, thus driving the fan blades **150** of the fan section **124**. In addition, or in the alternative, at least a portion of the electrical power **216** may be sent to the auxiliary electrical system **208** to support various electrical demands of the aircraft **10** (FIG. **1**).

[0033] As previously mentioned herein, the fuel treatment system **300** for the hybrid gas-electric propulsion system **200** generally includes a fuel pre-treatment unit fluidly coupled to the fuel tank **202**, a sweeping gas generator, a multi-way valve, a recuperator, a partial oxidation reformer, a solid oxide fuel cell, and a combustor, generally arranged as illustrated in the figures and as described herein. FIG. **4** is a schematic diagram of the fuel treatment system **300** for the hybrid gas-electric propulsion system **200** as shown in FIG. **3**, according to exemplary embodiments of the present disclosure.

[0034] As shown in FIG. **4**, the fuel treatment system **300** includes a fuel pre-treatment unit **302**. The fuel pre-treatment unit **302** incudes a fuel inlet **304** in fluid communication with fuel tank **202** and a fuel outlet **306**. A fuel flow passage **308** (shown in dashed lines) is fluidly coupled to the fuel inlet **304** and the fuel outlet **306** and provides a fluid flow path through the fuel pre-treatment unit **302**. The fuel flow passage **308** may be at least partially defined by various pipes, valves, fluid couplings and the like (not shown). As previously provided, the fuel pre-treatment unit **302** is configured to receive the hydrocarbon fuel **500** from the fuel tank **202** via fuel pump **204**. The fuel pump **204** provides a motive force to move the hydrocarbon fuel **500** from the fuel tank **202** through the fuel treatment system **300** via various pipes. valves, fluid couplings and the like (not shown). In exemplary embodiments. a fuel filter **218** may be disposed downstream from the fuel pump **204** and upstream from the fuel inlet **304** of the fuel pre-treatment unit **302** to remove particulates from the hydrocarbon fuel **500**.

[0035] In exemplary embodiments, the fuel pre-treatment unit **302** includes at least one heat exchanger for heating the hydrocarbon fuel **500**. In exemplary embodiments, the fuel pre-treatment unit **302** includes a pre-beat heat exchanger **310** in thermal communication with the hydrocarbon fuel **500** via fuel inlet **304**, and a fuel deoxygenation unit **312** in fluid and chemical communication with the hydrocarbon fuel **500** via the fuel flow passage **308**. The pre-heat heat exchanger **310** and the fuel deoxygenation unit **312** may be formed as a unitary device or body or may be formed from two separate bodies.

[0036] In exemplary embodiments, the pre-heat heat exchanger **310** is in thermal communication with a bleed air system **220**. The bleed air system **220** may include a hot-side fluid source **222**. The hot-side fluid source **222** may include either or both of the LP compressor **132** and the HP compressor **134** which provide a bot-side fluid (HSF), such as compressor bleed air, to the pre-heat heat exchanger **310** via various pipes, valves, fluid couplings and the like (not shown) that extend through the pre-heat heat exchanger **310**. A flow valve **224** may be disposed downstream from the hot-side fluid source **222** and upstream from the pre-heat heat exchanger **310**. The flow valve **224** may be configured and adjusted to control a flow rate of the hot-side fluid HSF provided to the pre-heat heat exchanger **310** and thus control the amount of heat transfer between the hot-side fluid HSF and the hydrocarbon fuel **500**. A downstream end of the bleed air system **220** may be fluidly connected to one or more of the HP turbine **138**, the HP compressor **134**, the environmental control system **24** for the aircraft **10** (FIG. **1**), and the cabin **22** of the aircraft **10** (FIG. **1**).

[0037] In exemplary embodiments, such as shown in FIG. **3**, the fuel treatment system **300** includes a sweeping gas generator **314** for providing a sweeping gas (SG) such as Nitrogen (N2) or other gas suitable for absorbing or sweeping oxygen from the hydrocarbon fuel **500**. The sweeping gas generator **314** is fluidly coupled to a sweeping gas circuit **316** formed at least in part by various pipes, valves, fluid couplings, and the like (not shown). The sweeping gas circuit **316** is fluidly coupled to and in fluid and chemical communication with the fuel deoxygenation unit **312** and is fluidly coupled to and in fluid communication with a partial oxidation reformer **318**. The partial oxidation reformer **318** includes a heated fuel inlet **320**, a reformed fuel outlet **322**. and an oxygen inlet **324**. In exemplary embodiments, the partial oxidation reformer **318** may be either be a Catalytic Partial Oxidation Reformer (preferred due to lower operation temp. −800 C) or Thermal Partial Oxidation Reformer (with operating temp. of ˜1200 C).

[0038] In exemplary embodiments, the sweeping gas generator **314** includes a sweeping gas and oxygen separation membrane, herein referred to as "SG-O2 separation membrane **326**". The sweeping gas generator **314** further includes an exhaust oxygen outlet **328** fluidly coupled to and in fluid communication with the oxygen inlet **324** of the partial oxidation reformer **318**. The exhaust oxygen outlet **328** is in fluid communication with the SG-O2 separation membrane **326**. The sweeping gas circuit **316** may include a pump **330** fluidly coupled to and in fluid communication with the sweeping gas generator **314** via a sweeping gas outlet **332** of the sweeping gas generator **314** and via a sweeping gas inlet **334** of the sweeping gas generator **314**.

[0039] The sweeping gas circuit **316** may include a flame arrester **336** disposed downstream from the fuel deoxygenation unit **312** and upstream from the sweeping gas inlet **334**. The sweeping gas generator **314** may be fluidly coupled to and in fluid communication with an ambient air source **338** such as, but not limited to. the nacelle **160** or the LP compressor **132**. A valve **340** may be fluidly coupled to and in fluid communication with the exhaust oxygen outlet **328** and the oxygen inlet **324** of the partial oxidation reformer **318**. In exemplary embodiments, the exhaust oxygen outlet **328** of the sweeping gas generator **314** is fluidly coupled to and in fluid communication with one or more of the environmental control system **24**, the oxygen storage tank **26**, and the vent **28** of the aircraft **10** (FIG. **1**) upstream of the valve **340**.

[0040] As shown in FIG. **4**, a multi-way valve **342** includes an inlet **344** that is fluidly coupled to and in fluid communication with the fuel tank **202** and the fuel flow passage **308** via the fuel outlet **306** of the fuel pre-treatment unit **302**. Actuation or manipulation of oxygen flow or fuel flow through either or both valve **340** and the multi-way valve **342** respectively may be controlled by a controller **346**. In at least certain exemplary embodiments, the controller **346** may be a full authority digital engine control ("FADEC") controller. However, in other embodiments, other suitable controllers may be provided. For example, in other embodiments, the controller **346** may include health monitoring units and other electronic systems.

[0041] The multi-way valve **342** includes at least a first outlet **348** and a second outlet **350**. The second outlet **350** may be fluidly coupled to and in fluid communication with a combustor **352** to enable direct fuel combustion during engine cold start-up. Combustor **352** may be integrated with combustion section **136** of the turbofan engine **120** as shown in FIG. **3** or may be independent thereof. For example. the combustor **352** may be part of a combustion system for an auxiliary power unit (not shown).

[0042] In exemplary embodiments, a heat exchanger or "recuperator **354**" including a first fuel passage **356** and a second fuel passage **358** is disposed downstream from the first outlet **348** of the multi-way valve **342**. More particularly, the first fuel passage **356** is fluidly coupled to and in fluid communication with the fuel pre-treatment unit **302** via the fuel outlet **306** and the first outlet **348** of the multi-way valve **342**. In operation, the first fuel passage **356** is in thermal communication with the second fuel passage **358** within the recuperator **354**. In exemplary embodiments. the recuperator **354** may be a liquid fuel-to-gas heat exchanger or the like. For example, the first fuel passage **356** may be configured to receive a liquid fuel such as the hydrocarbon fuel **500** while the second fuel passage may be configured to receive a gaseous fuel such as a hydrogen rich fuel.

[0043] In an exemplary embodiment, the fuel treatment system **300** includes a cold-start heat exchanger **360**. The cold-start heat exchanger **360** may be configured as fuel-to-gas heat exchanger or as an electric heater heat exchanger. The cold-start heat exchanger **360** includes fuel pre-heat passage **362** disposed upstream from the first fuel passage **356** of the recuperator **354**, in fluid communication with the fuel outlet **306** of the fuel pre-treatment unit **302**, and in fluid communication with the heated fuel inlet **320** of the partial oxidation reformer **318**. In particular embodiments, the fuel pre-heat passage **362** may be in fluid communication with the fuel outlet **306** of the fuel pre-treatment unit **302** via the first outlet **348** of the multi-way valve **342**.

[0044] The cold-start heat exchanger **360** may include an exhaust gas passage **364**. In operation, the fuel pre-heat passage **362** is in thermal communication with the exhaust gas passage **364**. The exhaust gas passage **364** may be fluidly coupled to and in fluid communication with the turbofan engine **120** (FIG. **3**). For example, the exhaust gas passage **364** may be in fluid communication with the LP compressor **132**, the HP compressor **134**, the combustion section **136**, the HP turbine **138**, the LP turbine, or the jet exhaust nozzle section **142** of the turbofan engine **120**. The cold-start heat exchanger **360**, when configured as an electric heater, may include a resistance or other suitable electric powered heater in thermal communication with the fuel pre-heat passage **362**.

[0045] In exemplary embodiments, the controller **346** is electronically connected to the cold-start heat exchanger **360** and is configured to control or allow fuel flow through the fuel pre-heat passage **362** of the cold-start heat exchanger **360**, thus bypassing or at least partially bypassing the first fuel passage **356** of the recuperator **354** in certain operating modes such as during a cold start of the turbofan engine **120**.

[0046] A temperature sensor **366** may be electronically coupled to the controller **346** and configured to measure and send an electronic signal indicative of fuel temperature as measured upstream from heated fuel inlet **320** of the partial oxidation reformer **318**. The controller **346** may be configured to adjust or manipulate either or both valve **340** and multi-way valve **342** from fully closed or zero flow positions, to fully open or full flow positions, or to any open positions defined therebetween, to meter or control a flow volume through the valve **340** or multi-way valve **342** respectively. As such, controller **346** may be configured to regulate the amount of fuel used to generate electric power vs. amount of fuel directly combusted. In addition, or in the alternative, controller **346** may be configured to control flow through the exhaust gas passage **364** or electricity provided to an electric heater of the cold-start heat exchanger **360**.

[0047] As shown in FIG. **4**, a solid oxide fuel cell **368** or “SOFC” is disposed downstream from the reformed fuel outlet **322** of the partial oxidation reformer **318**. The solid oxide fuel cell **368** includes an anode inlet **370**, an anode outlet **372**, a cathode inlet **374**, and a cathode outlet **376**. An anode flow passage **378** passes through the solid oxide fuel cell **368** and is fluidly coupled to and in fluid communication with the anode inlet **370** and the anode outlet **372**. A cathode flow passage **380** passes through the solid oxide fuel cell **368** and is fluidly coupled to and in fluid communication with the cathode inlet **374** and the cathode outlet **376**.

[0048] The anode inlet **370** is fluidly coupled to and in fluid communication with the reformed fuel outlet **322** of the partial oxidation reformer **318**. In exemplary embodiments, the anode outlet **372** may be in fluid communication with the second fuel passage **358** of the recuperator **354**. In addition, or in the alternative, the anode outlet **372** may be in fluid communication with the second fuel passage **358** of the recuperator **354**. In particular embodiments, the solid oxide fuel cell **368** is configured as a water-gas shift reactor to convert CO into CO2 and generate additional H2 fuel using water produced by the solid oxide fuel cell **368**.

[0049] In certain embodiments, a fuel and water separation unit **382** is disposed between the solid oxide fuel cell **368** and the recuperator **354** and in fluid communication with the anode outlet **372**. The fuel and water separation unit **382** includes a hydrogen fuel outlet **384** and an anode water outlet **386**. The hydrogen fuel outlet **384** is fluidly coupled to the second fuel passage **358** of the recuperator **354** and to the combustor **352**. In certain embodiments, the anode water outlet **386** may be fluidly coupled to and in fluid communication with one or more of the partial oxidation reformer

318, the HP turbine **138**, and the combustor **352**.

[0050] In particular embodiments, the solid oxide fuel cell **368** may be in thermal communication with a cathode air heating system **388** including a cathode air source **390** and an air-to-air heat exchanger **392** which may be configured as a recuperator. The cathode inlet **374** and the cathode outlet **376** are fluidly coupled to the cathode air heating system **388**. The cathode air source **390** may include the LP compressor **132** or the HP compressor **134**. The air-to-air heat exchanger **392** includes a cathode air passage **394** fluidly coupled to and in fluid communication with the cathode air source **390** and the cathode inlet **374**, and a cathode air exhaust passage **396** fluidly coupled to and in fluid communication with the cathode outlet **376**. In operation, cathode air passage **394** is in thermal communication with the cathode air exhaust passage **396**. In exemplary embodiments, the cathode air exhaust passage **396** may be fluidly coupled to and in fluid communication with a portion of the turbofan engine **120** (FIG. **3**) including but not limited to the HP compressor **134** and combustor **352**.

[0051] In exemplary embodiments, the cathode air heating system **388** may also include an auxiliary heater **398** such as an air-to-exhaust gas heat exchanger or electric heater. The auxiliary heater **398** is fluidly connected to and in fluid communication with the cathode air source **390** and the cathode inlet **374**. The auxiliary heater **398** may be fluidly connected and configured within the cathode air heating system **388** so as to bypass the air-to-air heat exchanger **392** during certain operating modes of the turbofan engine **120** (FIG. **2**) such as during a cold-start of the turbofan engine **120** (FIG. **2**). The auxiliary heater **398** includes a bypass air passage **400** in fluid communication with the cathode air source **390** and the cathode inlet **374**. In embodiments where the auxiliary heater **398** is configured as an air-to-exhaust gas heat exchanger, the auxiliary heater **398** includes an exhaust heat air passage **402** in fluid communication with an exhaust air source **404**. The exhaust air source **404** may include the turbofan engine **120** (FIG. **1**) such as but not limited to the jet exhaust nozzle section **142** (FIG. **3**). In embodiments where the auxiliary heater **398** is configured as an electric heater, the auxiliary heater **398** may include a resistance or other suitable electric powered heater in thermal communication with the bypass air passage **400**.

[0052] FIG. **5** is a schematic diagram of a portion of the fuel treatment system **300** for the hybrid gas-electric propulsion system **200** as shown in FIG. **4** with the addition of a post-heat heat exchanger **406** added to the fuel pre-treatment unit **302**, according to exemplary embodiments of the present disclosure. It is to be appreciated that the fuel treatment system **300** of FIGS. **4** and **5** have like components and use the same reference numbers for the common components. In the exemplary embodiment shown in FIG. **5**, the fuel pre-treatment unit **302** is in fluid communication with fuel tank **202** and is configured to receive the hydrocarbon fuel **500** therefrom. The pre-heat heat exchanger **310** is in thermal communication with the hydrocarbon fuel **500** via the fuel flow passage **308**. The fuel deoxygenation unit **312** is in fluid and chemical communication with the hydrocarbon fuel **500** downstream from the pre-heat heat exchanger **310**.

[0053] The post-heat heat exchanger **406** is disposed downstream from the fuel deoxygenation unit **312** and is in thermal communication with the hydrocarbon fuel **500** after it passes through the fuel deoxygenation unit **312**. The post-heat heat exchanger **406** is also fluidly connected to and in fluid communication with the inlet **344** of the multi-way valve **342** via the fuel outlet **306** of the fuel pre-treatment unit **302**. The pre-heat heat exchanger **310**, the fuel deoxygenation unit **312**, and the post-heat heat exchanger **406** may be formed or packaged as a unitary device or body. In the alternative, the pre-heat heat exchanger **310**, the fuel deoxygenation unit **312**, and the post-heat heat exchanger **406** may be formed from two or more separate devices.

[0054] The post-heat heat exchanger **406** is also in thermal communication with a hot-side fluid system **408**. The hot-side fluid system **408** may include an oil reservoir **410**, a pump **412**, and a heat source **414** such as a gearbox or bearing compartment. In operation, oil **416** is drawn from the oil reservoir **410** by the pump **412** and passed through the heat source **414** to pick up heat. The now heated oil **418** is routed through the post-heat heat exchanger **406** where at least a portion of the heat picked up by the oil **418** from the heat source **414** is transferred to the flow of the hydrocarbon fuel **500** flowing from the fuel deoxygenation unit **312**, thus cooling the heated oil **418** upstream from the oil reservoir **410**.

[0055] The fuel treatment system **300** as arranged and described herein and as shown in FIGS. **4** and **5** collectively, provides a system for processing the hydrocarbon fuel **500** and converting it into a hydrogen H2 fuel for a hybrid-electric aircraft by incorporating solid-oxide fuel cell with a partial oxidation reformer while addressing or preventing potential coke formation in the hydrocarbon fuel **500** at the high operating temperatures needed by the efficient electric power generation system of a hybrid electric aircraft.

[0056] Referring first to FIG. **4**, in operation, the hydrocarbon fuel **500** is drawn from the fuel tank **202**, passed through the fuel filter **218** (when present) to remove particulates generally present in the hydrocarbon fuel **500**, and then passed into the pre-heat heat exchanger **310** of the fuel pre-treatment unit **302**. Heat is transferred from the hot-side fluid HSF of the bleed air system **220** to the hydrocarbon fuel **500**. The hot-side fluid HSF is cooled thus producing a cooled hot-side fluid (CHSF) which may be routed to one or more of the environmental control system **24** which supply the cabin **22**, directly to the cabin **22**, the HP turbine **138**, and the HP compressor **134**. When the CHSF is routed to the environmental control system (ECS) **24**, the pre-heat heat exchanger **310** acts as the ECS precooler, eliminating the need for a separate ECS precooler heat exchanger and lowering engine weight. When the CHSF is routed to the HP turbine **138** and/or HP compressor **134**, it serves as a cooling air to enable engine operation at higher temperatures, which increases fuel efficiency. Flow valve **224** (when present) may be adjusted to control the flow rate of the hot-side fluid HSF through pre-heat heat exchanger **310** thus controlling the temperature of the hydrocarbon fuel **500** as it passes through the pre-heat heat exchanger **310** and into the fuel deoxygenation unit **312**.

[0057] As now heated hydrocarbon fuel **502** passes through the fuel deoxygenation unit **312**, the sweeping gas SG from the sweeping gas generator **314** flows through the sweeping gas circuit **316** and into the fuel deoxygenation unit **312** where it reacts with the heated hydrocarbon fuel **502** to sweep or absorb oxygen from the heated hydrocarbon fuel **502**, thus producing an oxygenated sweeping gas (OSG) and a heated deoxygenated hydrocarbon fuel **504**. The oxygenated sweeping gas OSG is routed back to the sweeping gas generator **314** wherein the SG-O2 separation membrane **326** separates the absorbed oxygen from the sweeping gas SG. The sweeping gas SG is allowed to flow back through the sweeping gas circuit **316** for continued operation. In an exemplary embodiment, the sweeping gas is Nitrogen (N2).

[0058] In particular embodiments, the SG-O2 separation membrane **326** may separate nitrogen (N2) from oxygen (O2) and other gases from ambient air (A) drawn into the sweeping gas generator **314** from the ambient air source **338**. The nitrogen may be used as the sweeping gas SG. This configuration allows for the generation of sweeping gas SG during operation of the aircraft **10** without having to carry a sweeping gas storage system onboard the aircraft **10** or onboard the turbofan engine **120**, thus saving weight and additional system components.

[0059] Oxygen (O2) separated from the oxygenated sweeping gas OSG or excess oxygen from ambient air A may be routed from the exhaust oxygen outlet **328** to the partial oxidation reformer **318**. In addition, or in the alternative, the oxygen O2 may be routed to one or more of the environmental control system **24**, the oxygen storage tank **26**, and the vent **28** for venting the oxygen O2 to atmosphere.

[0060] Referring briefly to FIG. **5**, in an exemplary embodiment wherein the post-heat heat exchanger **406** is present in the fuel pre-treatment unit **302** the now heated deoxygenated hydrocarbon fuel **504** may be routed through the post-heat heat exchanger **406** wherein additional heat is transferred from the heated oil **418** of the hot-side fluid system **408** to the heated deoxygenated hydrocarbon fuel **504**, thus further raising an operating temperature of the deoxygenated hydrocarbon fuel **500** upstream from the multi-way valve **342**.

[0061] Referring back to FIG. **4**, the controller **346** may send an electronic signal to the multi-way valve **342** to manipulate the multi-way valve **342** so as to control a flow rate of the heated deoxygenated hydrocarbon fuel **504** to one or more of the combustor **352** for combustion, to the recuperator **354** for further processing, and to the cold-start heat exchanger **360** for when the turbofan engine **120** is in a cold-start condition. The temperature sensor **366** may send a signal back to the controller **346** that is indicative of a temperature of the heated deoxygenated hydrocarbon fuel **504** just upstream from heated fuel inlet **320** of the partial oxidation reformer **318**.
[0062] In exemplary embodiments, if the temperature of the heated deoxygenated hydrocarbon fuel **504** is less than 800 C for when the partial oxidation reformer **318** is a catalytic partial oxidation reformer, or less than 1200 C for when the partial oxidation reformer **318** is a thermal partial oxidation reformer, the controller **346** may send an electronic signal to the cold-start heat exchanger **360** instructing it to open or otherwise receive at least a portion or all of the heated deoxygenated hydrocarbon fuel **504** from the first outlet **348** of the multi-way valve **342** for additional heating to the proper operational temperature of the partial oxidation reformer **318**. In a non-cold-start condition, the heated deoxygenated hydrocarbon fuel **504** is routed through the first fuel passage **356** of the recuperator **354** to provide the heated deoxygenated hydrocarbon fuel **504** to the partial oxidation reformer **318** at the appropriate temperature for operation.
[0063] As the heated deoxygenated hydrocarbon fuel **504** flows through the partial oxidation reformer **318** it reacts with the oxygen O2 from the exhaust oxygen outlet **328** of the sweeping gas generator **314**. The exothermic reaction produces a hydrogen rich+carbon monoxide fuel mixture or "H2+CO fuel mixture **506**". The hot H2+CO fuel mixture **506** is then routed into the anode inlet **370** of the solid oxide fuel cell **368**. Within the solid oxide fuel cell, the H2+CO fuel mixture **506** reacts as an anode with cathode air (CA) provided by the cathode air source **390** of the cathode air heating system **388**. The cathode air CA reacts as a cathode within the solid oxide fuel cell **368**. Excess air (EA) from the exothermic reaction within the solid oxide fuel cell **368** may be routed from the cathode outlet **376**, through the air-to-air heat exchanger **392** and on to the one or more other components or systems of the turbofan engine **120** or the aircraft **10** such as but not limited to the HP compressor **134** or a combustion chamber.
[0064] In exemplary embodiments, the solid oxide fuel cell **368** acts as a water-gas shift reactor to the reaction in the solid oxide fuel cell **368** between the H2+CO fuel mixture **506** and the cathode air CA, thus converting the CO into CO2 and generating more hydrogen fuel using water produced by the solid oxide fuel cell **368**. This results in a hydrogen+carbon dioxide and water fuel mixture or "H2+CO2+H2O fuel mixture **508**". In embodiments, the fuel and water separator **382** may separate the H2+CO2 from the H2O, thus producing a hydrogen and carbon dioxide fuel mixture or "hot H2+CO2 fuel mixture **510**" and anode water **512**.
[0065] Because the reactions in the partial oxidation reformer **318** and the solid oxide fuel cell **368** are both exothermic, the hot H2+CO2 fuel mixture **510** may be routed to the recuperator **354** to preheat the heated deoxygenated hydrocarbon fuel **504**, before hot H2+CO2 fuel mixture **510** is then routed into the combustor **352** for combustion. The hot H2+CO2 fuel mixture **510** may be routed to the recuperator **354** for additional heating upstream from the combustor **352** and then routed into the combustor **352** for combustion. The anode water **512** may be routed from the fuel and water separation unit **382** to one or more of the HP turbine **138** for cooling, the combustor for NOx reduction, and the partial oxidation reformer **318** to reduce soot formation therein, thus increasing life expectancy of a catalyst of the partial oxidation reformer **318**.
[0066] This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they

include equivalent structural elements with insubstantial differences from the literal languages of the claims.

[0067] Further aspects are provided by the subject matter of the following clauses:

[0068] A fuel treatment system for a hybrid gas-electric propulsion system using a hydrocarbon fuel, the fuel treatment system comprising: a fuel pre-treatment unit including a fuel inlet, a fuel outlet, and a fuel flow passage defined therebetween; a recuperator including a first fuel passage and a second fuel passage, wherein the first fuel passage is in fluid communication with the fuel outlet of the fuel pre-treatment unit; a partial oxidation reformer including a heated fuel inlet and a reformed fuel outlet, wherein the heated fuel inlet is in fluid communication with the fuel pre-treatment unit via the first fuel passage; a solid oxide fuel cell including an anode inlet and an anode outlet, wherein the anode inlet is in fluid communication with the reformed fuel outlet of the partial oxidation reformer, and wherein the anode outlet is in fluid communication with the second fuel passage of the recuperator; and a combustor in fluid communication with the anode outlet via the second fuel passage.

[0069] The fuel treatment system of the previous or any following clause, wherein the fuel pre-treatment unit includes a pre-heat heat exchanger and a fuel deoxygenation unit, wherein the fuel flow passage passes through the pre-heat heat exchanger and the fuel deoxygenation unit.

[0070] The fuel treatment system of any previous or following clause, wherein the fuel pre-treatment unit includes a post-heat heat exchanger disposed downstream from and in fluid communication with the fuel deoxygenation unit via the fuel flow passage.

[0071] The fuel treatment system of any previous or following clause, further comprising a sweeping gas generator including a sweeping gas circuit, wherein the fuel pre-treatment unit includes a fuel deoxygenation unit, wherein the sweeping gas circuit is in fluid communication with the fuel deoxygenation unit.

[0072] The fuel treatment system of any previous or following clause, wherein the sweeping gas generator includes a sweeping gas and oxygen separation membrane.

[0073] The fuel treatment system of any previous or following clause, wherein the partial oxidation reformer includes an oxygen inlet, wherein the sweeping gas generator includes an exhaust oxygen outlet in fluid communication with the sweeping gas and oxygen separation membrane, wherein the exhaust oxygen outlet is in fluid communication with the oxygen inlet of the partial oxidation reformer.

[0074] The fuel treatment system of any previous or following clause, wherein the solid oxide fuel cell is configured as a water-gas shift reactor.

[0075] The fuel treatment system of any previous or following clause, further comprising a cold-start heat exchanger including a fuel pre-heat passage, wherein the fuel pre-heat passage is in fluid communication with the fuel outlet of the fuel pre-treatment unit and the heated fuel inlet of the partial oxidation reformer.

[0076] The fuel treatment system of any previous or following clause, wherein the fuel pre-heat passage is in fluid communication with the fuel outlet of the fuel pre-treatment unit upstream from the first fuel passage of the recuperator.

[0077] The fuel treatment system of any previous or following clause, wherein the cold-start heat exchanger includes an exhaust gas passage in fluid communication with a turbofan engine, wherein the fuel pre-heat passage is in thermal communication with the exhaust gas passage.

[0078] The fuel treatment system of any previous or following clause, wherein the cold-start heat exchanger includes an electric heater, wherein the fuel pre-heat passage is in thermal communication with the electric heater.

[0079] The fuel treatment system of any previous or following clause, further comprising a fuel and water separation unit in fluid communication with the anode outlet, wherein the fuel and water separation unit includes a hydrogen fuel outlet in fluid communication with the combustor via the second fuel passage of the recuperator.

[0080] The fuel treatment system of any previous or following clause, wherein the fuel and water separation unit includes an anode water outlet, wherein the anode water outlet is in fluid communication with the partial oxidation reformer.
[0081] The fuel treatment system of any previous or following clause, wherein the fuel and water separation unit includes an anode water outlet, wherein the anode water outlet is in fluid communication with the combustor.
[0082] The fuel treatment system of any previous or following clause, wherein the fuel and water separation unit includes an anode water outlet, wherein the anode water outlet is in fluid communication with a high pressure turbine of a turbofan engine.
[0083] The fuel treatment system of any previous or following clause, further comprising a controller, wherein the controller is configured to regulate one or more of an oxygen flowrate, fuel flowrate, and fuel temperature supplied to the partial oxidation reformer.
[0084] The fuel treatment system of any previous or following clause, further comprising a cathode air heating system, wherein the solid oxide fuel cell includes a cathode inlet and a cathode outlet, wherein the cathode inlet and the cathode outlet are fluidly coupled to the cathode air heating system.
[0085] The fuel treatment system of any previous or following clause, wherein the cathode air heating system includes a cathode air source and an air-to-air heat exchanger, wherein the air-to-air heat exchanger includes a cathode air passage in fluid communication with the cathode inlet, and a cathode air exhaust passage in fluid communication with the cathode outlet.
[0086] The fuel treatment system of any previous or following clause, wherein the cathode air heating system includes a cathode air source and an auxiliary heater, wherein the auxiliary heater includes a bypass air passage in fluid communication with a cathode air source and the cathode inlet, and an exhaust heat air passage in fluid communication with an exhaust air source.
[0087] The fuel treatment system of any previous or following clause, wherein the cathode air heating system includes a cathode air source and an auxiliary heater, wherein the auxiliary heater includes a bypass air passage in fluid communication with a cathode air source and the cathode inlet, and an electric heater in thermal communication with the bypass air passage.

Claims

1. A fuel treatment system for a hybrid gas-electric propulsion system using a hydrocarbon fuel, the fuel treatment system comprising: a fuel pre-treatment unit including a fuel inlet, a fuel outlet, and a fuel flow passage defined therebetween; a recuperator including a first fuel passage and a second fuel passage, wherein the first fuel passage is in fluid communication with the fuel outlet of the fuel pre-treatment unit; a partial oxidation reformer including a heated fuel inlet and a reformed fuel outlet, wherein the heated fuel inlet is in fluid communication with the fuel pre-treatment unit via the first fuel passage; a solid oxide fuel cell including an anode inlet and an anode outlet, wherein the anode inlet is in fluid communication with the reformed fuel outlet of the partial oxidation reformer, and wherein the anode outlet is in fluid communication with the second fuel passage of the recuperator; and a combustor in fluid communication with the anode outlet via the second fuel passage.
2. The fuel treatment system of claim 1, wherein the fuel pre-treatment unit includes a pre-heat heat exchanger and a fuel deoxygenation unit, wherein the fuel flow passage passes through the pre-heat heat exchanger and the fuel deoxygenation unit.
3. The fuel treatment system of claim 2, wherein the fuel pre-treatment unit includes a post-heat heat exchanger disposed downstream from and in fluid communication with the fuel deoxygenation unit via the fuel flow passage.
4. The fuel treatment system of claim 1, further comprising a sweeping gas generator including a sweeping gas circuit, wherein the fuel pre-treatment unit includes a fuel deoxygenation unit,

wherein the sweeping gas circuit is in fluid communication with the fuel deoxygenation unit.
5. The fuel treatment system of claim 4, wherein the sweeping gas generator includes a sweeping gas and oxygen separation membrane.
6. The fuel treatment system of claim 5, wherein the partial oxidation reformer includes an oxygen inlet, wherein the sweeping gas generator includes an exhaust oxygen outlet in fluid communication with the sweeping gas and oxygen separation membrane, wherein the exhaust oxygen outlet is in fluid communication with the oxygen inlet of the partial oxidation reformer.
7. The fuel treatment system of claim 1, wherein the solid oxide fuel cell is configured as a water-gas shift reactor.
8. The fuel treatment system of claim 1, further comprising a cold-start heat exchanger including a fuel pre-heat passage, wherein the fuel pre-heat passage is in fluid communication with the fuel outlet of the fuel pre-treatment unit and the heated fuel inlet of the partial oxidation reformer.
9. The fuel treatment system of claim 8, wherein the fuel pre-heat passage is in fluid communication with the fuel outlet of the fuel pre-treatment unit upstream from the first fuel passage of the recuperator.
10. The fuel treatment system of claim 8, wherein the cold-start heat exchanger includes an exhaust gas passage in fluid communication with a turbofan engine, wherein the fuel pre-heat passage is in thermal communication with the exhaust gas passage.
11. The fuel treatment system of claim 8, wherein the cold-start heat exchanger includes an electric heater, wherein the fuel pre-heat passage is in thermal communication with the electric heater.
12. The fuel treatment system of claim 1, further comprising a fuel and water separation unit in fluid communication with the anode outlet, wherein the fuel and water separation unit includes a hydrogen fuel outlet in fluid communication with the combustor via the second fuel passage of the recuperator.
13. The fuel treatment system of claim 12, wherein the fuel and water separation unit includes an anode water outlet, wherein the anode water outlet is in fluid communication with the partial oxidation reformer.
14. The fuel treatment system of claim 12, wherein the fuel and water separation unit includes an anode water outlet, wherein the anode water outlet is in fluid communication with the combustor.
15. The fuel treatment system of claim 12, wherein the fuel and water separation unit includes an anode water outlet, wherein the anode water outlet is in fluid communication with a high pressure turbine of a turbofan engine.
16. The fuel treatment system of claim 1, further comprising a controller, wherein the controller is configured to regulate one or more of an oxygen flowrate, fuel flowrate, and fuel temperature supplied to the partial oxidation reformer.
17. The fuel treatment system of claim 1, further comprising a cathode air heating system, wherein the solid oxide fuel cell includes a cathode inlet and a cathode outlet, wherein the cathode inlet and the cathode outlet are fluidly coupled to the cathode air heating system.
18. The fuel treatment system of claim 17, wherein the cathode air heating system includes a cathode air source and an air-to-air heat exchanger, wherein the air-to-air heat exchanger includes a cathode air passage in fluid communication with the cathode inlet, and a cathode air exhaust passage in fluid communication with the cathode outlet.
19. The fuel treatment system of claim 17, wherein the cathode air heating system includes a cathode air source and an auxiliary heater, wherein the auxiliary heater includes a bypass air passage in fluid communication with a cathode air source and the cathode inlet, and an exhaust heat air passage in fluid communication with an exhaust air source.
20. The fuel treatment system of claim 17, wherein the cathode air heating system includes a cathode air source and an auxiliary heater, wherein the auxiliary heater includes a bypass air passage in fluid communication with a cathode air source and the cathode inlet, and an electric heater in thermal communication with the bypass air passage.